CALLUM GREIG

MECHATRONICS DESIGN ENGINEER

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PROFFESSIONAL OVERVIEW

Mechatronics design engineer with strong CAD modelling and design intuition, supported by analysis of composite structures, embedded systems, and automation hardware. Experienced in material/structural analysis, full stack development of internal applications/tools, and embedded instrumentation development.

WORK EXPERIENCE

Mechatronics Design Engineer

Gracol Composites | *Nov 2023 – Present*

- Collaborated with senior engineer on design of composite structural systems, applying mechanical analysis, material characterization, and standardized testing to validate structural performance.
- Designed 3D-printed forming systems using parametric surface modelling to improve pultrusion efficiency and reduce stress concentrations during forming.
- Developed embedded monitoring systems for manufacturing processes, including pultrusion load monitoring, VOC sensing, resin exotherm thermal analysis, NDT monitoring, and automated dangerous goods temperature monitoring with alert notifications.
- Developed internal engineering tools including quoting calculators, structural beam analysis tools, automated pultrusion setup sheets, and production databases to improve workflow and manufacturing traceability.

Workshop Engineering Intern

Reed Industrial LTD | *Nov 2021 – Feb 2022*

KEY PROJECTS

CNC Machine

Gracol Composites | *2024 -2026*

- Lead Project Engineer – design, implementation, testing, and management.
- Designed and implemented a CNC control architecture integrating motion drives, power systems, sensors, machine I/O, tooling automation, integrated safety systems, and operational manuals/SOP.

Flexural Material Testing Machine

Gracol Composites | *2023 - 2024*

- Lead Project Engineer – design, implementation, programming, testing, and management.
- Designed a semi-automated three-point flexural testing machine with embedded control, load cell and displacement instrumentation, signal conditioning, and stress–strain analysis for composites to ASTM D790.

Full Year Project – Non-Destructive Testing (NDT) of Composite Structures

University of Canterbury | *Jan 2023 – Dec 2023*

- Lead NDT analyst, research, and design.
- NDT analysis of composite airframe, evaluate validity of NDT methods for detecting CFRP long-term performance, airframe load testing, and evaluate real-time diagnosis methods of airframe structural health.

EDUCATION

Bachelor of Mechatronics Engineering

University of Canterbury | 2023

TECHNICAL SKILLS

- **CAD & Analysis:** Parametric CAD, FEA, CFD, PCB Design
- **Programming:** C, C++, Python
- **Embedded & Control Systems:** PLC Ladder Logic, G-Code, Microcontrollers
- **Systems & Software Tools:** Git, Bash, SQL, Docker, Proxmox (VMs), YAML
- **Web & Scripting:** JavaScript, HTML, VBA
- **Sales & Systems Engineering:** Pricing & Structural Calculators, Quoting, Management, and Financial Planning

PERSONAL PROJECTS / ACHIEVEMENTS

- **Airfoil Disc Golf Project (2024)**

Applied aerodynamic principles to design, and 3D print TPU disc golf discs. Iterating geometry based on flight performance and manufacturability.

- **Multi-Protocol IoT Device Design (Present)**

Embedded multi-protocol device design integrating RF, NFC, digital I/O, and inertial sensing.

- **NASA Space Camp (2018)**

Participated in team-based aerospace mission simulation and engineering design activities.

REFERENCE

Mathew Bancroft

Operations Manager | Gracol

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